IS 3920 – Individual Project on Business Solutions

Software Requirements Specification

for

RailPost

Digital Cargo Tracking and Management System for Railway Stations

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Student's Declaration

I hereby declare that this report is my original work and has not been submitted, in whole or in part, for any degree or diploma at any university or other institution of higher learning. All information derived from the work of others, whether published or unpublished, has been properly acknowledged in the text, and a complete list of references has been provided.

2026.01.19…………………… ……………………………………
Date Signature of Student

Supervisors' declaration

I hereby declare that I have reviewed this project and find it adequate in both scope and quality.

1. Name of Supervisor: Dr. Leelanga D. Seneviratne

   Designation: Senior Lecturer, Head of Department (IDS)

   Date:19.01.2026

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   Designation: Senior Tech Lead, Zafer AI

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Revision History

Name	Date	Reason for Changes	Version
E.R.T. Wickramasinghe	13.01.2026	Initial SRS Creation	1.0

Chapter 1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to comprehensively describe the functional and nonfunctional requirements of the RailPost – Digital Cargo Tracking and Management System. This document acts as a formal agreement between stakeholders and developers, clearly defining system behavior, constraints, interfaces, and quality expectations. It also serves as a reference throughout system development, testing, validation, and future maintenance.

1.2 Document Conventions

This document adheres to the IEEE 830 / IEEE 29148 Software Requirements Specification standards. Functional requirements are denoted using the term “shall” to indicate mandatory behavior, while “should” indicates recommended features. Diagrams and tables are used where appropriate to enhance clarity.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Academic supervisors and examiners
- Software developers and system designers
- Railway administrators and operational staff
- Future maintenance and enhancement teams

Readers are advised to begin with Sections 1 and 2 for an overall understanding of the system, followed by Sections 3 and 4 for detailed requirements.

1.4 Product Scope

RailPost is a centralized digital cargo management solution designed specifically for Sri Lanka Railways. The system replaces inefficient manual cargo handling processes with an automated, transparent, and reliable digital workflow that supports real-time tracking, notifications, and accurate record-keeping across railway stations.

1.4.1 Problem in brief

The existing railway cargo service in Sri Lanka heavily relies on paper forms, handwritten receipts, and physical registers. This results in delays, data inconsistencies, misplaced cargo, inaccurate cost calculations, and frequent disputes. Senders and receivers lack visibility into cargo status, while station staff face increased administrative burdens and operational inefficiencies.

1.4.2 Aim and Objectives

Aim:

To design and implement a digital cargo tracking and management system that improves operational efficiency, transparency, accountability, and trust in railway cargo services.

Objectives:

- Digitize cargo booking and registration processes
- Provide real-time cargo tracking and notifications
- Maintain accurate and centralized cargo records
- Automate transport and insurance cost calculations
- Generate digital receipts and delivery confirmations
- Reduce disputes and enhance customer satisfaction

1.5 References

1.IEEE 29148:2018 – Systems and Software Engineering – Life Cycle Processes – Requirements Engineering

2.Sri Lanka Railways Reservation

3.Ministry of Transport

Chapter 2 Overall description

This section provides a comprehensive overview of the RailPost system, describing how it fits within the existing railway cargo ecosystem, the key functions it delivers, the different user classes involved, and the assumptions and constraints under which the system is designed and implemented. The purpose of this section is to give stakeholders a high-level understanding of the system before examining detailed functional requirements.

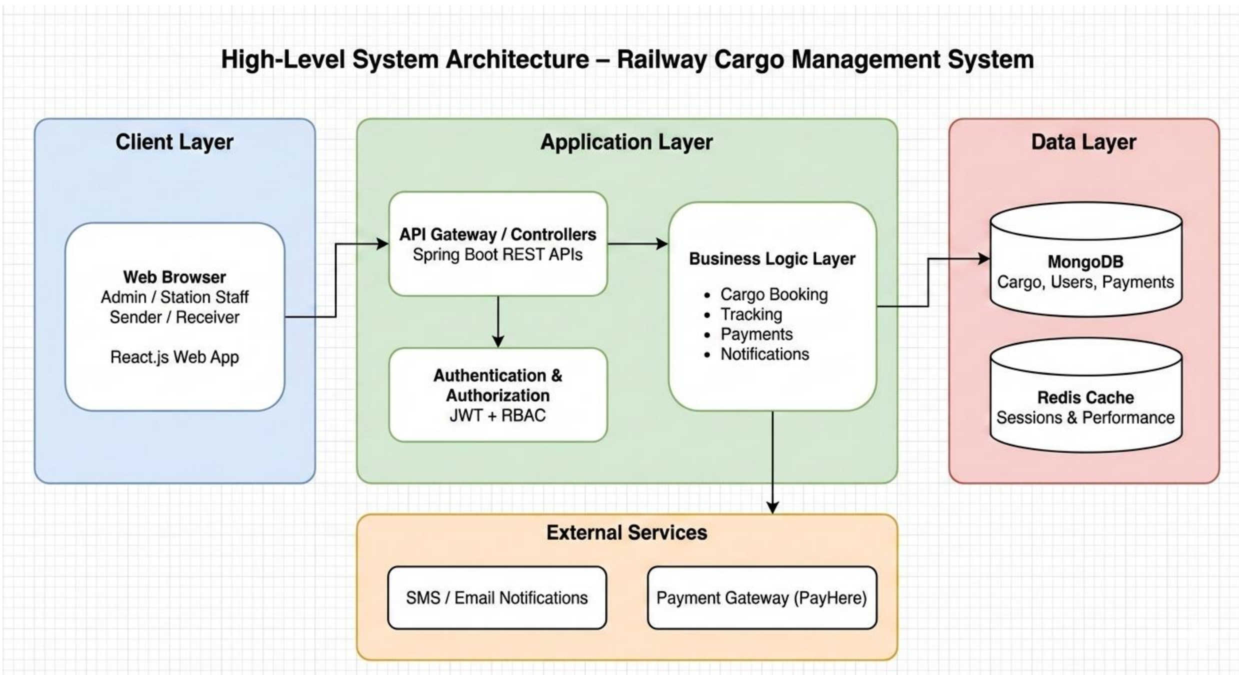

Figure 01- System Architecture Diagram

2.1 Product Perspective

RailPost is a web-based and mobile-supported system integrated into the existing railway cargo operation. It acts as a centralized platform connecting senders, receivers, station officers, station masters, and administrators.

The system architecture follows a client–server model with secure APIs and a centralized database.

2.2 Product Functions

Major system functions include:

- User authentication and role-based access control for all system roles
- Online cargo booking by registered senders
- Walk-in cargo registration by station officers on behalf of senders
- Automated transport and insurance cost calculation based on predefined rules
- Generation of unique tracking numbers and QR codes for each cargo item
- Scanning of QR codes at dispatch, transit, and arrival points
- Real-time cargo status updates and timeline visualization
- Automated notification delivery via SMS and Email
- OTP-based verification for secure cargo pickup
- Digital receipt generation for booking and delivery
- Administrative configuration of stations, users, and cost matrices
- Generation of operational and management reports

These functions collectively aim to reduce manual effort, eliminate data inconsistencies, and provide transparency to all stakeholders.

2.3 User Classes and Characteristics

RailPost supports multiple user classes, each with distinct responsibilities, access levels, and usage patterns.

Administrator: Administrators are responsible for system governance and configuration. They are typically IT or railway management personnel with advanced technical skills. Administrators use the system infrequently but perform critical tasks such as creating railway stations, assigning station masters, configuring cost matrices, and viewing system-wide reports. Administrators do not participate in operational cargo handling.

Station Master: Station masters supervise cargo operations at individual railway stations. They have moderate technical proficiency and use the system daily to monitor station-level activities, manage station officers, and generate operational reports. Their role focuses on oversight rather than direct data entry.

Station Officer: Station officers are frontline operational users who interact with the system frequently throughout the day. They handle walk-in cargo bookings, scan QR codes during dispatch and arrival, and verify OTPs during delivery. The system interface for this user class must be simple, efficient, and error tolerant.

Sender: Senders include individuals and businesses who use the railway cargo service. Their technical skills vary widely. Some senders use the system online, while others rely on station officers for registration. Senders primarily use the system to book cargo, track shipments, and access receipts.

Receiver: Receivers are end recipients of cargo. They typically do not require full system access or accounts. Receivers interact with the system mainly through tracking links and SMS notifications and provide OTPs during cargo pickup.

Reviewer: Reviewers are third parties who are granted read-only access to cargo tracking information by senders. This user class may include family members, business partners, or supervisors who need visibility but no control over cargo operations.

2.4 Operating Environment

RailPost is designed to operate in a distributed railway environment with varying levels of infrastructure quality. At railway stations, the system is accessed through standard desktop computers or laptops equipped with modern web browsers. Mobile devices may also be used by station officers for QR code scanning.

The system requires a stable internet connection to communicate with centralized backend services. However, it is designed with tolerance for intermittent connectivity, which may occur at rural or remote stations. The backend is hosted on a secure server environment capable of handling concurrent users across multiple stations.

RailPost supports common web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. The user interface is responsive and compatible with mobile devices to accommodate different usage contexts.

2.5 Design and Implementation Constraints

The design and implementation of RailPost are subject to several constraints. Budgetary limitations restrict the use of proprietary software and require reliance on open-source or widely available technologies. Time constraints associated with academic project timelines also influence development scope and prioritization.

As the system is intended for use within a government-controlled railway environment, it must align with institutional policies, approval processes, and operational practices. Training time for station staff is limited, requiring the system to be intuitive and easy to learn.

Technology constraints include the need to support existing hardware at railway stations and operate reliably under varying network conditions.

2.6 User Documentation

RailPost will be supported by comprehensive user documentation tailored to different user classes. This includes user manuals for administrators and station masters, quick-reference guides for station officers, and simplified instructions for senders and receivers.

Documentation will include step-by-step instructions, screenshots, and explanations of common tasks. Online help features and FAQs will be provided within the system to assist users during operation. Where necessary, documentation may be provided in multiple languages to improve accessibility.

2.7 Assumptions and Dependencies

The development and operation of RailPost are based on several assumptions. It is assumed that railway stations have access to basic computing hardware and internet connectivity. It is also assumed that station staff are willing to adopt the new system and participate in training activities.

The system depends on external services such as SMS and email gateways for notification delivery. Reliable operation of these services is critical to meeting communication requirements. RailPost also depends on accurate configuration of cost and distance data by administrators to ensure correct calculations.

Chapter 3 External Interface Requirements

3.1 User Interfaces

RailPost provides a unified and intuitive user interface designed to accommodate users with varying levels of technical expertise. The system features a single-entry login page for all authenticated roles, after which users are redirected to role-specific dashboards.

Each dashboard presents relevant information such as cargo summaries, alerts, and actionable tasks. The sender dashboard focuses on booking and tracking, while station officer dashboards emphasize scanning, status updates, and delivery processing. The admin interface provides configuration panels and analytical views.

Receivers and reviewers access cargo tracking through a simplified tracking page without login requirements. The user interface is fully responsive and optimized for both desktop and mobile devices.

3.2 Hardware Interfaces

RailPost interacts with minimal physical hardware to ensure ease of deployment. The primary hardware interface is QR code scanning, performed using mobile device cameras. Optional hardware includes:

- Printers for generating physical receipts
- Digital weighing scales (manual weight entry supported)

The system does not require specialized or proprietary hardware.

3.3 Software Interfaces

The system interfaces with several external software services:

- SMS gateway services for notification delivery
- Email servers for electronic communication
- Authentication services for secure access control
- Payment gateways for transaction processing (future enhancement)

All integrations are handled through secure APIs

3.4 Communication Interfaces

RailPost uses standard web communication protocols. All data exchanges occur over HTTPS using RESTful APIs and JSON data formats. Encryption is enforced to protect sensitive data during transmission. Error handling mechanisms ensure reliable communication even during temporary network failures**.**

Chapter 4 System Features

This section describes the major system features of the RailPost system in detail. Each feature is presented with its description, priority, actors involved, preconditions, main processing flow, alternative flows, and postconditions. These features collectively define the functional behavior of the system and address the problems identified in the existing cargo handling process.

4.1 Online Cargo Booking

4.1.1 Description and Priority

This feature enables registered senders to book cargo online through the RailPost web or mobile application. It replaces the traditional paper-based booking process and ensures accurate, complete, and validated data entry. The system captures cargo, sender, and receiver details and automatically calculates transportation and insurance costs based on predefined rules.

Priority: High

User Role: Sender (Registered Customer)

4.1.2 Stimulus/Response Sequences

Stimulus: A registered sender logs into the RailPost system and initiates a new cargo booking by entering cargo details, receiver information, and destination station.

Response: The system validates all entered data, calculates transportation and insurance costs based on weight, category, distance, and declared value, displays the total cost to the sender, processes payment upon confirmation, generates a unique tracking number and QR code, creates a digital receipt, and sends notifications to both sender and receiver.

Stimulus: A sender enters invalid or incomplete cargo details during the booking process.

Response: The system displays inline validation errors highlighting the problematic fields, provides clear error messages explaining what corrections are needed, prevents form submission until all errors are resolved, and retains correctly entered data to avoid reentry.

Stimulus: Payment processing fails during booking confirmation due to network issues or insufficient funds.

Response: The system displays an error message explaining the payment failure reason, saves the booking as a draft with pending payment status, provides options to retry payment or use an alternative payment method, and does not generate tracking number or QR code until payment succeeds.

4.1.3 Functional Requirements

REQ 1.1: The system shall provide a booking form accessible only to authenticated registered senders.

REQ 1.2: The system shall require mandatory fields including cargo weight, category, dimensions, declared value, receiver name, receiver contact number, and destination station.

REQ 1.3: The system shall validate cargo weight to be within acceptable limits (minimum 1 kg, maximum 1000 kg for standard booking).

REQ 1.4: The system shall provide a dropdown list of predefined cargo categories including Fragile, Perishable, Documents, Electronics, General Goods, and Hazardous Materials.

REQ 1.5: The system shall validate receiver contact number to be a valid 10 digit mobile number.

REQ 1.6: The system shall provide a searchable dropdown of all active destination stations with autocomplete functionality.

REQ 1.7: The system shall calculate transportation cost based on weight, distance, and cargo category using the configured rate matrix.

REQ 1.8: The system shall calculate insurance cost as a percentage of declared value (default 2 percent, configurable by admin).

REQ 1.9: The system shall display itemized cost breakdown showing base transportation cost, insurance cost, applicable taxes, and total amount.

REQ 1.10: The system shall support multiple payment methods including credit/debit card, digital wallet, net banking, and cash on registration (for walk in).

REQ 1.11: The system shall integrate with payment gateway for processing online payments securely.

REQ 1.12: The system shall generate a unique alphanumeric tracking number (format: RP YYYYMMDD XXXXXX) upon successful payment.

REQ 1.13: The system shall generate a QR code encoding the tracking number, sender ID, receiver details, and destination station.

REQ 1.14: The system shall create a digital receipt in PDF format containing booking details, payment information, and QR code.

REQ 1.15: The system shall send booking confirmation SMS to sender's registered mobile number.

REQ 1.16: The system shall send booking confirmation email to sender's registered email address with attached receipt.

REQ 1.17: The system shall send notification SMS to receiver informing about incoming cargo with tracking number.

REQ 1.18: The system shall allow senders to save booking as draft and complete it later before payment.

REQ 1.19: The system shall display estimated delivery date based on destination and current operational schedules.

REQ 1.20: The system shall maintain audit trail of all booking transactions with timestamps and user IDs.

REQ 1.21: The system shall prevent duplicate bookings by validating against recent submissions from the same user.

REQ 1.22: The system shall provide booking history view showing all past and current bookings for the logged in sender.

4.2 Walk-in Cargo Registration

4.2.1 Description and Priority

This feature supports walk in senders who visit railway stations physically to dispatch cargo. Station officers assist in registering cargo details into the system on behalf of senders, ensuring that even non digital users benefit from the system. This feature maintains inclusivity while digitalizing the cargo handling process.

Priority: High

User Role: Station Officer, Sender (Walk in Customer)

4.2.2 Stimulus/Response Sequences

Stimulus: A walk-in sender arrives at the railway station with cargo and approaches the station officer to register the shipment.

Response: The station officer logs into the RailPost system, navigates to the walk in registration module, enters sender and receiver details provided by the customer, weighs the cargo using the station scale and enters the weight, selects cargo category and destination, system calculates costs automatically, officer collects payment (cash or card), system generates QR code and receipt, officer prints and provides receipt to sender, and notifications are sent to both sender and receiver.

Stimulus: A walk in sender does not have receiver's complete contact information.

Response: The system allows registration with partial receiver details but marks the booking as requiring verification, sends notification to station officer to follow up with sender, and prevents cargo dispatch until receiver details are completed or verified.

4.2.3 Functional Requirements

REQ 2.1: The system shall provide a walk in registration interface accessible only to authenticated station officers.

REQ 2.2: The system shall allow station officers to create bookings on behalf of senders without requiring sender to have a registered account.

REQ 2.3: The system shall capture walk in sender details including name, contact number, and optional email address.

REQ 2.4: The system shall validate walk in sender contact number to ensure it is a valid mobile number for SMS notifications.

REQ 2.5: The system shall integrate with digital weighing scale via USB or serial port to automatically capture cargo weight.

REQ 2.6: The system shall allow manual weight entry with mandatory officer confirmation if scale integration fails.

REQ 2.7: The system shall display real time cost calculation as cargo details are entered.

REQ 2.8: The system shall support cash payment recording with automatic change calculation.

REQ 2.9: The system shall support card payment through point of sale (POS) terminal integration.

REQ 2.10: The system shall generate and print QR code label immediately after payment confirmation.

REQ 2.11: The system shall print receipt on thermal printer or standard printer based on station configuration.

REQ 2.12: The system shall create a sender profile for walk-in customers to enable future tracking and bookings.

REQ 2.13: The system shall log the station officer ID who processed the walk in registration.

REQ 2.14: The system shall display queue management information showing number of pending registrations.

REQ 2.15: The system shall allow bulk registration for multiple cargo items from the same sender in a single transaction.

REQ 2.16: The system shall provide quick templates for common cargo types to speed up registration.

REQ 2.17: The system shall verify receiver contact number by sending test SMS during registration (optional).

REQ 2.18: The system shall maintain daily cash collection report for each station officer.

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4.3 QR Code Generation and Management

4.3.1 Description and Priority

This feature generates a unique QR code for each cargo item that serves as the primary identifier throughout the cargo journey. The QR code encodes essential cargo information and enables rapid scanning at checkpoints, reducing manual data entry errors and improving tracking accuracy.

Priority: High

User Role: System, Station Officer

4.3.2 Stimulus/Response Sequences

Stimulus: A cargo booking is confirmed and payment is successfully processed.

Response: The system automatically generates a unique QR code encoding tracking number, origin station, destination station, sender ID, receiver contact, cargo category, and booking timestamp, stores the QR code image in the database, associates it with the cargo record, includes it in the digital receipt, and makes it available for printing on cargo labels.

Stimulus: A station officer scans a cargo QR code using a handheld scanner at any checkpoint.

Response: The system decodes the QR code, retrieves the complete cargo record from the database, displays cargo details on the officer's screen, prompts for status update action (dispatch, arrival, delivery), updates cargo status with timestamp and location, and triggers notifications to relevant parties.

Stimulus: A QR code becomes damaged or unreadable during transit.

Response: The system allows station officers to search cargo by tracking number, regenerates and reprints the QR code, updates the cargo record with replacement QR code details, and logs the incident in the audit trail.

4.3.3 Functional Requirements

REQ 3.1: The system shall generate QR codes using industry standard QR code format (ISO/IEC 18004).

REQ 3.2: The system shall encode the following information in QR code: tracking number, origin station code, destination station code, sender reference ID, booking date, and cargo category.

REQ 3.3: The system shall use error correction level H (high) to ensure QR code remains readable even if up to 30 percent is damaged.

REQ 3.4: The system shall generate QR codes with minimum size of 2x2 cm for easy scanning.

REQ 3.5: The system shall store QR code images in PNG format with 300 DPI resolution.

REQ 3.6: The system shall print QR codes on adhesive labels (standard size 10x10 cm) with cargo details in human readable format.

REQ 3.7: The system shall support QR code scanning via handheld barcode scanners, mobile device cameras, and fixed mount scanners.

REQ 3.8: The system shall decode scanned QR codes within 2 seconds and retrieve cargo information.

REQ 3.9: The system shall validate scanned QR codes to ensure they belong to active cargo bookings.

REQ 3.10: The system shall log all QR code scan events with timestamp, location, and officer ID.

REQ 3.11: The system shall allow regeneration of QR codes for damaged labels with proper authorization.

REQ 3.12: The system shall prevent duplicate QR code generation for the same cargo booking.

REQ 3.13: The system shall include QR code verification checksum to detect tampering or counterfeit labels.

REQ 3.14: The system shall provide QR code printing template with railway logo and cargo handling instructions.

REQ 3.15: The system shall support batch QR code generation for multiple cargo items.

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4.4 Cargo Dispatch and Transit Tracking

4.4.1 Description and Priority

This feature tracks cargo movement from origin station through intermediate stations to the destination. Real-time status updates occur when cargo is dispatched, arrives at intermediate stations, and reaches the destination, providing complete visibility into the cargo journey.

Priority: High

User Role: Station Officer, System

4.4.2 Stimulus/Response Sequences

Stimulus: A station officer at the origin station scans cargo QR code before loading it onto the train for dispatch.

Response: The system retrieves cargo details, prompts officer to enter train number and scheduled departure time, updates cargo status to Dispatched with timestamp, records departure location and officer ID, sends SMS notification to sender confirming dispatch with train details, sends SMS notification to destination station alerting about incoming cargo, and updates the cargo tracking timeline.

Stimulus: A train carrying cargo arrives at an intermediate station and cargo is offloaded for transit storage.

Response: The system detects QR code scan at intermediate stations, updates cargo status to In Transit At (station name), records arrival timestamp and current location, calculates estimated time to destination, sends update notification to sender if tracking notifications are enabled, and maintains transit log for audit purposes.

Stimulus: Cargo arrives at the destination station and is scanned by the receiving officer.

Response: The system updates cargo status to Arrive at Destination with timestamp, calculates actual transit time and compares with estimated delivery date, sends SMS notification to receiver with pickup instructions and OTP, sends email notification to sender confirming safe arrival, flags any delayed shipments for supervisor review, and makes cargo available for delivery processing.

Stimulus: Cargo has been in transit for longer than the estimated delivery time without status update.

Response: The system automatically flags cargo as Delayed, generates alert notification to station supervisors, sends status inquiry to last known location, updates sender with delay notification and revised delivery estimate, and escalates to central operations if delay exceeds 24 hours.

4.4.3 Functional Requirements

REQ 4.1: The system shall provide dispatch interface for station officers to scan and process outgoing cargo.

REQ 4.2: The system shall require train number entry during dispatch to associate cargo with specific train.

REQ 4.3: The system shall validate train number against master train schedule database.

REQ 4.4: The system shall record exact dispatch timestamp automatically upon status update.

REQ 4.5: The system shall capture station officer ID who performed the dispatch operation.

REQ 4.6: The system shall update cargo status to Dispatched and make it visible in tracking system within 30 seconds.

REQ 4.7: The system shall send dispatch confirmation SMS to sender including train number and estimated arrival time.

REQ 4.8: The system shall send advance notification to destination station about incoming cargo.

REQ 4.9: The system shall support batch dispatch for multiple cargo items on the same train.

REQ 4.10: The system shall allow scanning cargo at intermediate stations to update transit progress.

REQ 4.11: The system shall automatically determine intermediate station location based on scan location and update status accordingly.

REQ 4.12: The system shall maintain complete transit history showing all scan points with timestamps.

REQ 4.13: The system shall calculate estimated time of arrival based on train schedule and current location.

REQ 4.14: The system shall provide real time tracking dashboard showing all cargo in transit.

REQ 4.15: The system shall flag cargo that has not been scanned within expected time windows.

REQ 4.16: The system shall send arrival notification to receiver when cargo reaches destination station.

REQ 4.17: The system shall generate daily dispatch reports for each station showing all dispatched cargo.

REQ 4.18: The system shall prevent dispatch of cargo without valid QR code and booking confirmation.

REQ 4.19: The system shall support manual status update with supervisor approval if QR scanning fails.

REQ 4.20: The system shall integrate with railway operational system to retrieve train schedules and route information.

4.5 Cargo Delivery and OTP Verification

4.5.1 Description and Priority

This feature ensures secure cargo delivery using One Time Password (OTP) verification to confirm receiver identity. This prevents unauthorized pickup and ensures cargo is delivered to the rightful recipient, addressing a major concern in traditional cargo handling.

Priority: High

User Role: Station Officer, Receiver

4.5.2 Stimulus/Response Sequences

Stimulus: A receiver arrives at the destination station to collect cargo and provides tracking number or shows notification SMS to the station officer.

Response: The station officer searches for the cargo in the system, verifies cargo status is Arrived at Destination, initiates delivery process, system generates a 6 digit OTP and sends it to receiver's registered mobile number via SMS, displays waiting screen for officer, and prompts receiver to provide the OTP.

Stimulus: Receiver provides the correct OTP to the station officer within the validity period (15 minutes).

Response: The system validates the OTP against the generated code, confirms successful verification, prompts officer to confirm cargo handover, updates cargo status to Delivered with delivery timestamp, records receiver signature or acknowledgment, sends delivery confirmation SMS to sender, generates delivery receipt, closes the cargo transaction, and updates delivery statistics.

Stimulus: Receiver provides an incorrect OTP or OTP has expired.

Response: The system displays error message indicating invalid or expired OTP, allows up to 3 retry attempts, provides option to resend new OTP to receiver's mobile, logs failed verification attempt, and requires supervisor authorization for delivery after 3 failed attempts.

Stimulus: Receiver's mobile phone is not available or not working at the time of pickup.

Response: The system allows alternative verification using receiver’s issued ID proof, requires station officer to manually verify ID details against booking records, requires supervisor approval for OTP bypass, captures photo of receiver's ID document, updates cargo record with alternative verification method used, and sends post-delivery notification to sender informing about alternative verification.

4.5.3 Functional Requirements

REQ 5.1: The system shall provide delivery interface for station officers to initiate cargo pickup process.

REQ 5.2: The system shall allow cargo search by tracking number, receiver mobile number, or receiver name.

REQ 5.3: The system shall verify cargo status is Arrived at Destination before allowing delivery initiation.

REQ 5.4: The system shall generate a random 6 digit numeric OTP for each delivery request.

REQ 5.5: The system shall send OTP to receiver's registered mobile number via SMS within 10 seconds.

REQ 5.6: The system shall set OTP validity period to 15 minutes from generation time.

REQ 5.7: The system shall display countdown timer showing OTP expiry time to the station officer.

REQ 5.8: The system shall validate entered OTP against generated OTP in real time.

REQ 5.9: The system shall allow maximum 3 OTP verification attempts before requiring supervisor intervention.

REQ 5.10: The system shall provide Resend OTP option with 60 second cooldown period between resends.

REQ 5.11: The system shall invalidate previous OTP when new OTP is generated for the same cargo.

REQ 5.12: The system shall update cargo status to Delivered immediately upon successful OTP verification.

REQ 5.13: The system shall record delivery timestamp, station officer ID, and verification method.

REQ 5.14: The system shall send delivery confirmation SMS to sender with delivery timestamp

REQ 5.15: The system shall support alternate verification using government ID proof with supervisor approval.

4.6 Notification Management

4.6.1 Description and Priority

This feature manages automated notifications via SMS and email to keep senders and receivers informed throughout the cargo journey. Timely notifications improve customer experience, reduce inquiry calls, and ensure transparency in cargo handling.

Priority: Medium

User Role: System, Sender, Receiver

4.6.2 Stimulus/Response Sequences

Stimulus: A cargo status changes to any of the defined states (Booked, Dispatched, In Transit, Arrived, Delivered).

Response: The system identifies the status change event, retrieves sender and receiver contact information from cargo record, selects appropriate notification template based on status and recipient type, personalizes message content with cargo specific details (tracking number, location, timestamp), sends SMS notification to configured mobile numbers, sends email notification to configured email addresses, logs notification delivery status, and retries failed notifications up to 3 times.

Stimulus: A cargo is delayed beyond the estimated delivery date.

Response: The system automatically detects delay by comparing current time with estimated delivery date, generates delay notification with reason and revised delivery estimate, sends SMS to both sender and receiver, escalates to station supervisor for investigation, and updates cargo record with delay flag and reason.

Stimulus: The notification SMS delivery fails due to network issues or invalid mobile number.

Response: The system logs the failed notification attempt with error reason, retries SMS delivery using alternate SMS gateway, stores undelivered notification in pending queue, alerts station officer to inform customer manually, and updates notification delivery status in cargo record.

4.6.3 Functional Requirements

REQ 6.1: The system shall send automated notifications for all major cargo status changes (Booked, Dispatched, Arrived, Delivered).

REQ 6.2: The system shall support both SMS and email notification channels.

REQ 6.3: The system shall use SMS for time critical notifications (OTP, arrival, ready for pickup).

REQ 6.4: The system shall use email for detailed notifications with attachments (receipt, delivery confirmation).

REQ 6.5: The system shall maintain notification templates for each cargo status with placeholders for dynamic content.

REQ 6.6: The system shall personalize notification messages with recipient name, tracking number, station name, and timestamp.

REQ 6.7: The system shall send booking confirmation notification immediately after payment success.

REQ 6.8: The system shall send dispatch notification within 5 minutes of cargo being loaded on train.

REQ 6.9: The system shall send arrival notification to receiver within 10 minutes of cargo reaching destination.

REQ 6.10: The system shall send delivery confirmation to sender within 5 minutes of successful handover.

REQ 6.11: The system shall integrate with SMS gateway provider for reliable SMS delivery.

REQ 6.12: The system shall integrate with email service (SMTP) for email notifications.

REQ 6.13: The system shall implement retry mechanism with exponential backoff for failed notifications.

REQ 6.14: The system shall log all notification attempts with delivery status, timestamp, and error details.

REQ 6.15: The system shall provide notification history view in cargo tracking interface.

REQ 6.16: The system shall allow users to configure notification preferences (SMS only, Email only, Both).

REQ 6.17: The system shall support multiple recipient mobile numbers for the same cargo (up to 3).

REQ 6.18: The system shall validate mobile numbers and email addresses before sending notifications.

REQ 6.19: The system shall generate daily notification delivery reports showing success/failure rates.

REQ 6.20: The system shall send delay notifications when cargo exceeds estimated delivery time by 6 hours.

REQ 6.21: The system shall send reminder notifications to receivers for uncollected cargo after 24 hours of arrival.

REQ 6.22: The system shall support notification language selection based on user preference (English, Regional languages).

REQ 6.23: The system shall include tracking URL in SMS notifications for web-based tracking.

REQ 6.24: The system shall maintain opt out list for users who request to stop notifications.

4.7 Real Time Cargo Tracking

4.7.1 Description

This feature provides real time cargo tracking interface for senders, receivers, and station officers to view current cargo status and complete journey history. It enhances transparency and reduces customer inquiries.

Priority: High

User Role: Sender, Receiver, Station Officer

4.7.2 Stimulus/Response Sequences

Stimulus: A sender or receiver enters tracking number in the tracking interface or clicks tracking link from notification SMS.

Response: The system validates tracking number format, retrieves cargo record from database, displays current cargo status with visual indicators, shows complete journey timeline with all scan events, displays estimated delivery date if cargo is in transit, shows destination station contact information, and provides options to download receipt or contact support.

4.7.3 Functional Requirements

REQ 7.1: The system shall provide public tracking interface accessible without login for basic tracking.

REQ 7.2: The system shall require only tracking number for public cargo status inquiry.

REQ 7.3: The system shall display cargo current status (Booked, Dispatched, In Transit, Arrived, Delivered).

REQ 7.4: The system shall show visual timeline of cargo journey with completed and pending stages.

REQ 7.5: The system shall display timestamp and location for each status update.

REQ 7.6: The system shall show estimated delivery date and time when cargo is in transit.

REQ 7.7: The system shall provide map view showing current cargo location and route.

REQ 7.8: The system shall display origin and destination station details with contact information.

REQ 7.9: The system shall show sender and receiver information (masked for privacy in public view).

REQ 7.10: The system shall provide detailed view of cargo specifications and declared value for authenticated users.

REQ 7.11: The system shall allow bulk tracking for multiple cargo items by uploading tracking number list.

REQ 7.12: The system shall provide tracking status API for integration with third party applications.

REQ 7.13: The system shall cache tracking data to improve response time for frequently queried cargo.

REQ 7.14: The system shall update tracking status in real time within 30 seconds of any status change.

4.8 Reporting and Analytics

4.8.1 Description and Priority

This feature provides comprehensive reports and analytics on cargo volumes, revenue, operational performance, and service quality metrics. Reports support data driven decision making and operational improvements.

Priority: Low

User Role: Admin, Station Master, Supervisor

4.8.2 Stimulus/Response Sequences

Stimulus: A station master logs into the system and requests monthly cargo volume report for their station.

Response: The system prompts for date range selection, station master selects month and year, system retrieves all cargo bookings for the selected period and station, aggregates data by cargo category, calculates total volume, revenue, and average transit time, generates report with charts and tables, displays report on screen with option to download as PDF or Excel, and saves report generation activity in audit log.

Stimulus: An administrator requests system wide revenue analysis for quarterly review.

Response: The system retrieves all completed bookings for the quarter, calculates total revenue by station, by cargo category, and by month, generates trend analysis charts, compares with previous quarter and same quarter last year, identifies top performing stations and categories, generates executive summary dashboard, and exports comprehensive report in PDF format.

4.8.3 Functional Requirements

REQ 8.1: The system shall provide reporting dashboard accessible to admin and station master roles.

REQ 8.2: The system shall generate daily cargo booking report showing all bookings with totals.

REQ 8.3: The system shall generate daily revenue collection report by station and by officer.

REQ 8.4: The system shall generate daily dispatch and delivery report showing cargo movement.

REQ 8.5: The system shall generate weekly cargo volume report by category and by destination.

REQ 8.6: The system shall generate monthly revenue report with comparison to previous months.

REQ 8.7: The system shall generate monthly performance report showing average transit time and delivery success rate.

REQ 8.8: The system shall generate quarterly trend analysis report with charts and graphs.

REQ 8.9: The system shall generate annual summary report for management review.

REQ 8.10: The system shall provide adhoc report generation with custom date range and filters.

REQ 8.11: The system shall allow report export in multiple formats (PDF, Excel, CSV).

REQ 8.12: The system shall provide visualization dashboards with charts (bar, line, pie) for key metrics.

REQ 8.13: The system shall show real time analytics dashboard with current day statistics.

REQ 8.14: The system shall calculate and display key performance indicators (KPIs): total bookings, revenue, average transit time, delivery success rate, customer satisfaction score.

REQ 8.15: The system shall provide station wise comparison report showing relative performance.

REQ 8.16: The system shall generate delayed cargo report flagging shipments exceeding delivery estimates.

REQ 8.17: The system shall generate unclaimed cargo report for shipments not picked up after 48 hours.

REQ 8.18: The system shall provide officer productivity report showing bookings processed per officer.

REQ 8.19: The system shall generate payment collection report with payment method breakdown.

REQ 8.20: The system shall provide customer analytics showing repeat customer rate and booking patterns.

REQ 8.21: The system shall allow report scheduling for automatic generation and email delivery.

REQ 8.22: The system shall maintain report generation history with timestamp and user details.

REQ 8.23: The system shall support role based report access control.

Chapter 5 Other Nonfunctional Requirements

5.1 Performance Requirements

The RailPost system shall meet the following performance criteria:

- The system shall respond to user actions (such as login, cargo booking, and status updates) within 3 seconds under normal operating conditions.
- QR code scanning and cargo status updates shall be reflected in the system within 5 seconds.
- The system shall support concurrent usage by multiple users across different railway stations without performance degradation.
- The system shall be capable of handling peak loads during high cargo volume periods such as festive seasons.

5.2 Safety Requirements

Although RailPost does not directly control physical railway operations, it supports safety by ensuring accurate and reliable information handling.

- The system shall prevent duplicate or inconsistent cargo records.
- The system shall provide confirmation messages for critical actions such as cargo dispatch and delivery.
- Regular automated data backups shall be performed to prevent data loss.
- In the event of system failure, previously stored data shall remain intact and recoverable.

5.3 Security Requirements

Security is critical due to the handling of sensitive cargo and user information.

- The system shall enforce role-based access control to restrict unauthorized actions.
- All user authentications shall be handled using secure mechanisms such as JWT-based authentication.
- User passwords shall be stored in encrypted and hashed formats.
- All data transmission shall occur over encrypted HTTPS connections.
- OTP verification shall be mandatory for cargo delivery to prevent unauthorized pickup.
- System activity logs shall be maintained for audit purposes.

5.4 Software Quality Attributes

The RailPost system shall adhere to the following quality attributes:

Reliability: The system shall operate continuously with minimal downtime and recover quickly from failures.

Usability: The user interface shall be simple and intuitive, enabling users with basic digital literacy to operate the system with minimal training.

Scalability: The system architecture shall support future expansion, including the addition of new railway stations and increased cargo volume.

Maintainability: The system should be modular and well-documented to facilitate future enhancements and maintenance.

Availability: The system shall be available at least 99% of the time, excluding scheduled maintenance.

5.5 Business Rules

The RailPost system shall enforce the following business rules:

- Every cargo item has a unique tracking number and QR code.
- Cargo status shall be updated at every key stage of transportation.
- Delivery shall not be completed without successful OTP verification.
- Admin users shall not perform operational cargo handling tasks.
- Cost and insurance calculations shall follow predefined rules configured by the administrator.

5.6 Legal and Regulatory Requirements

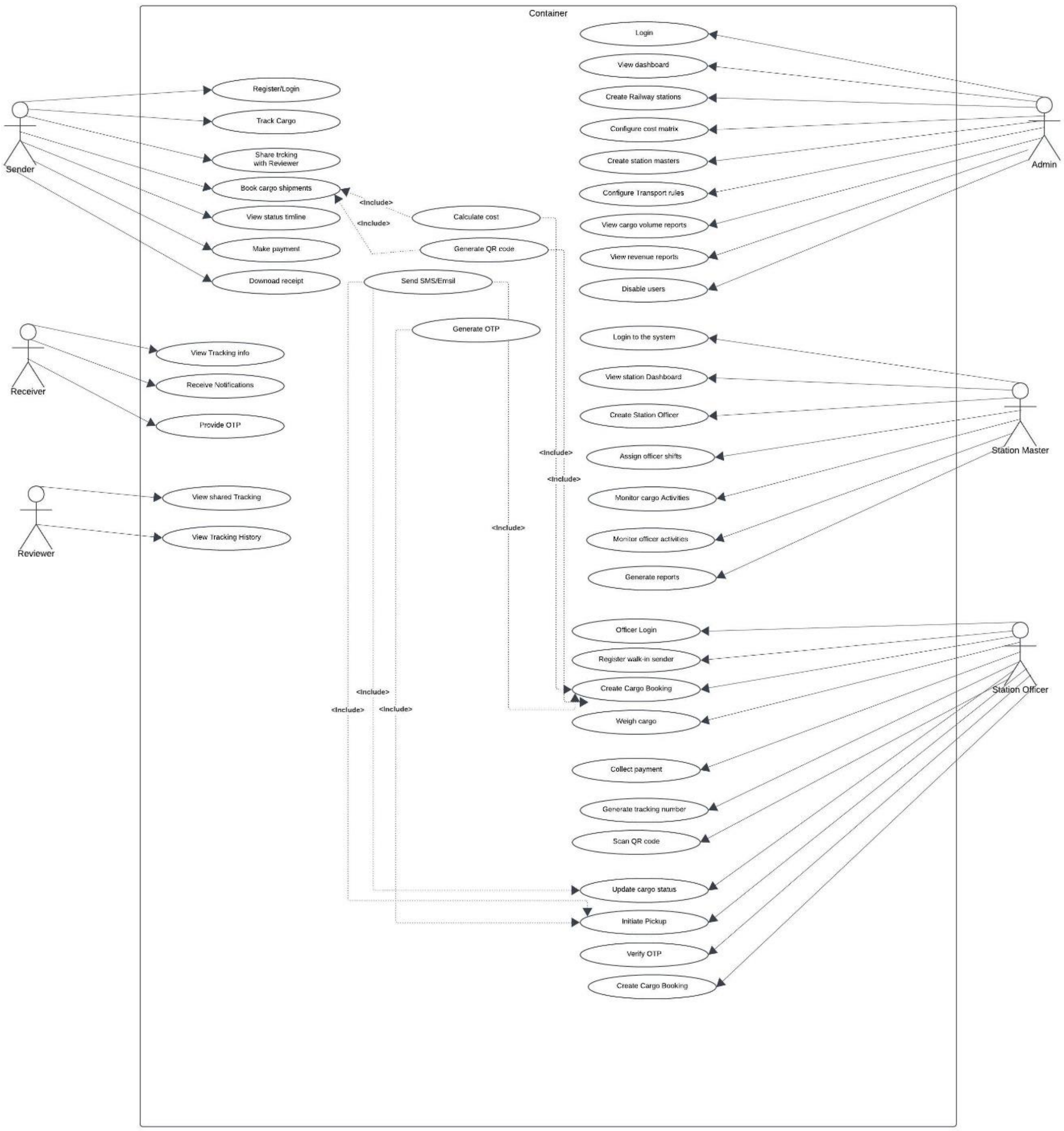
- The system shall comply with Sri Lanka Railways operational policies.
- Personal data handling shall comply with applicable data protection regulations
- Transaction records shall be retained for auditing and reporting purposes.

Appendix A – Glossary

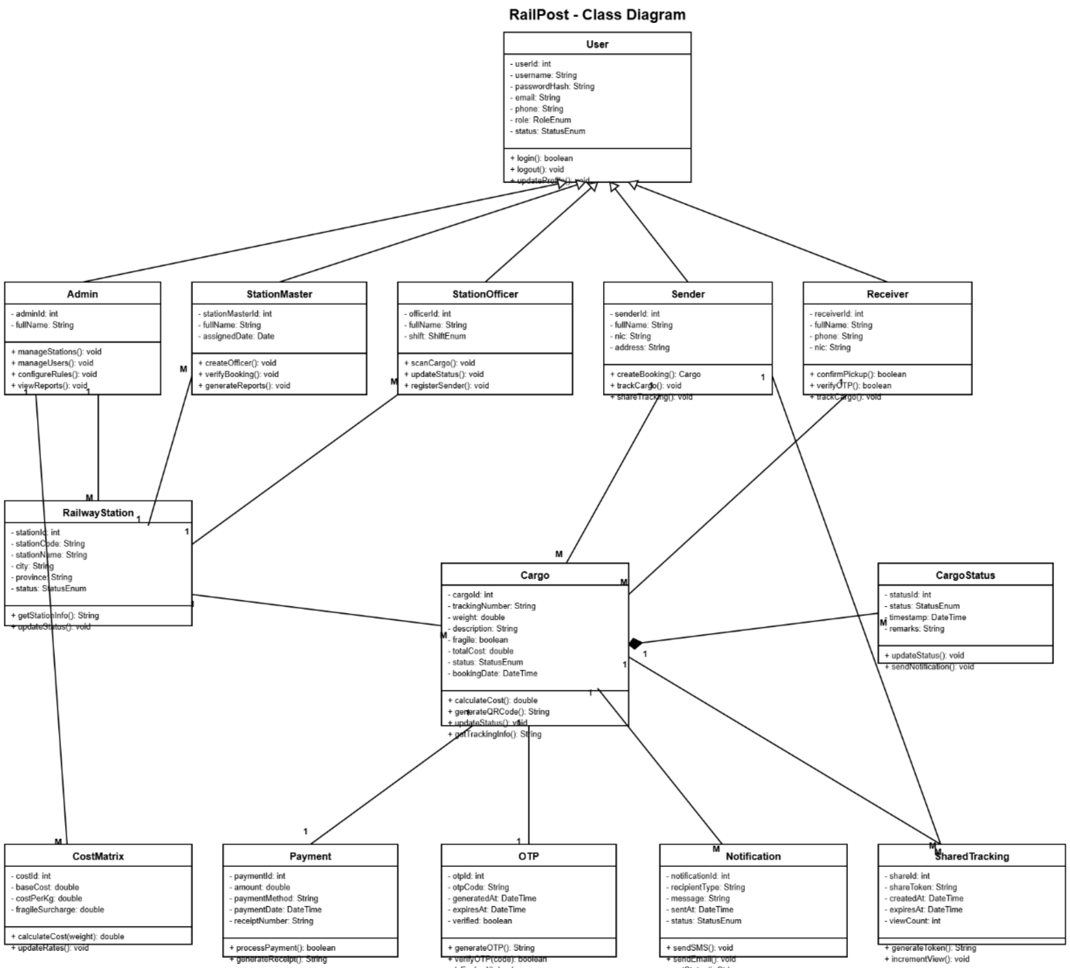
- QR Code: Machine-readable code for identification
- OTP: One-Time Password

Appendix B – Analysis Models

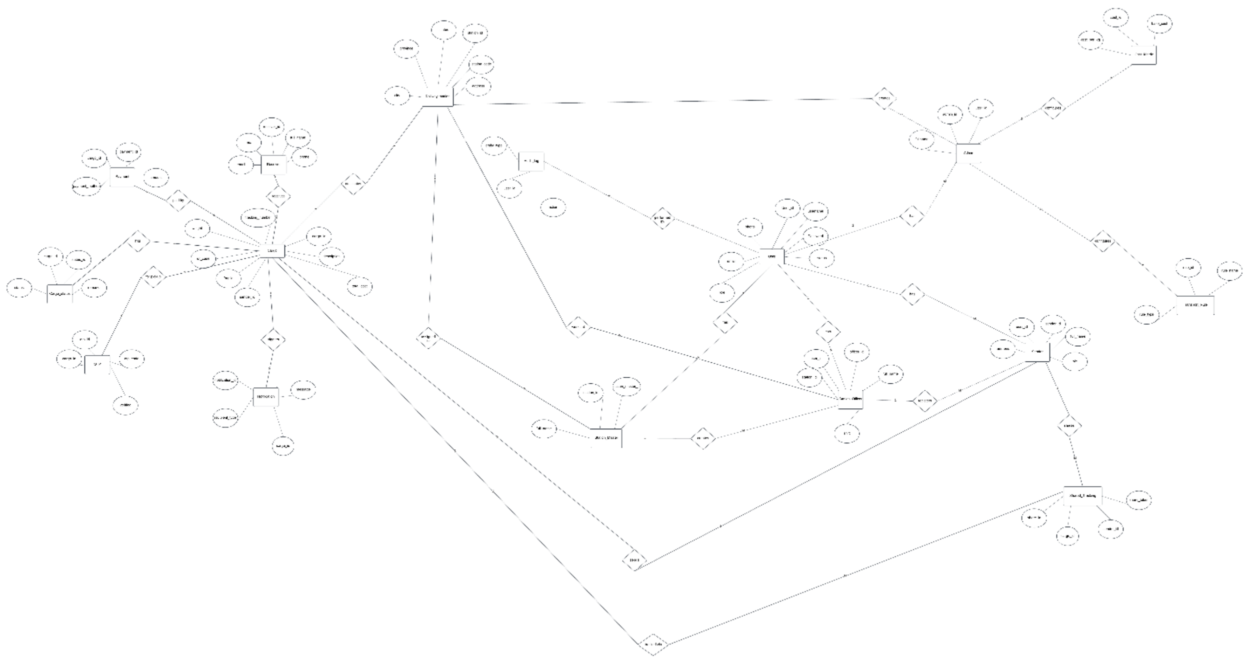
Use Case Diagram



Class Diagram



ER Diagram



Appendix C – UI Designs

Figure 02 – Home dashboard (Login for all users-RBAC)

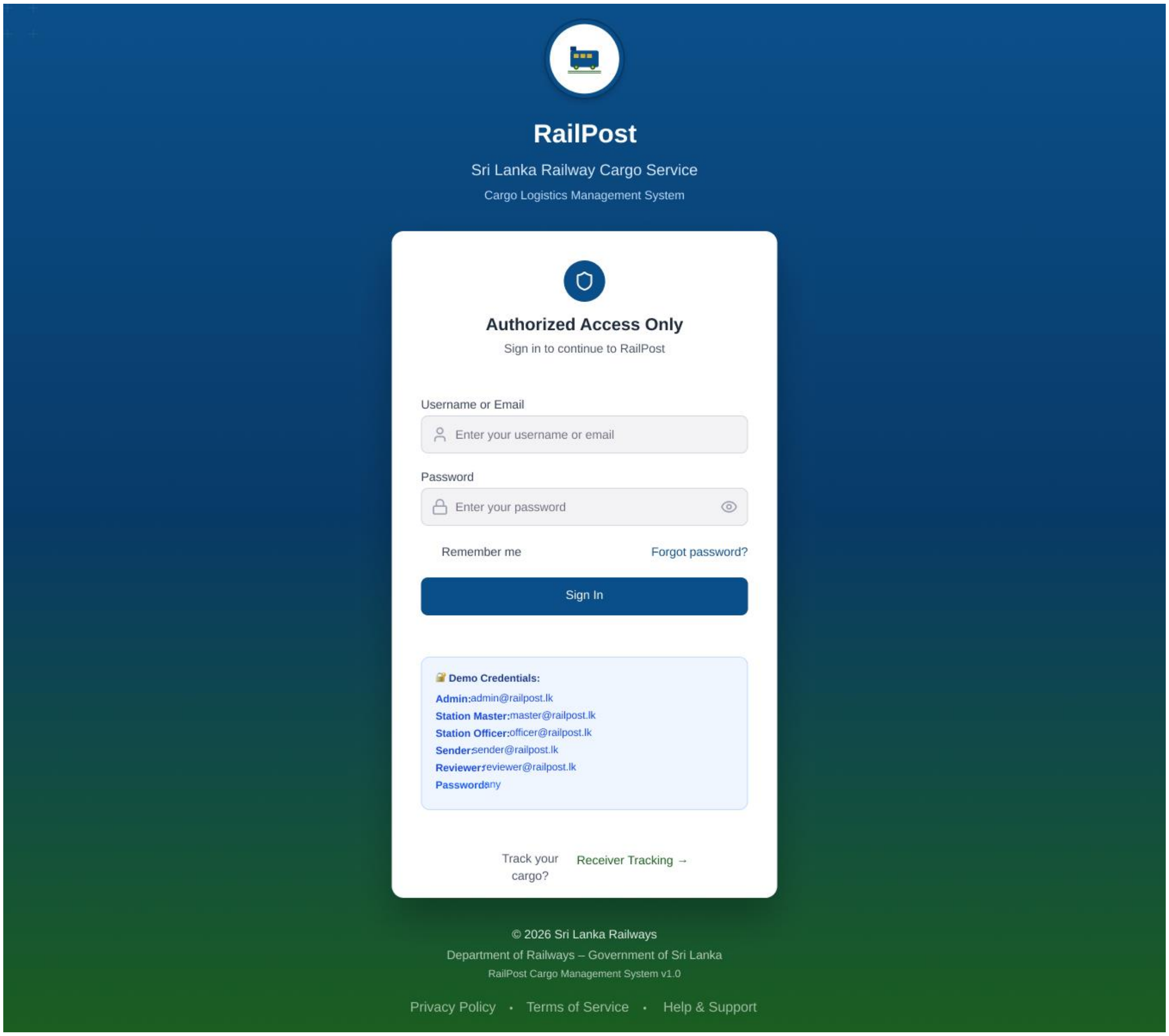


Figure 03 – Admin Dashboard

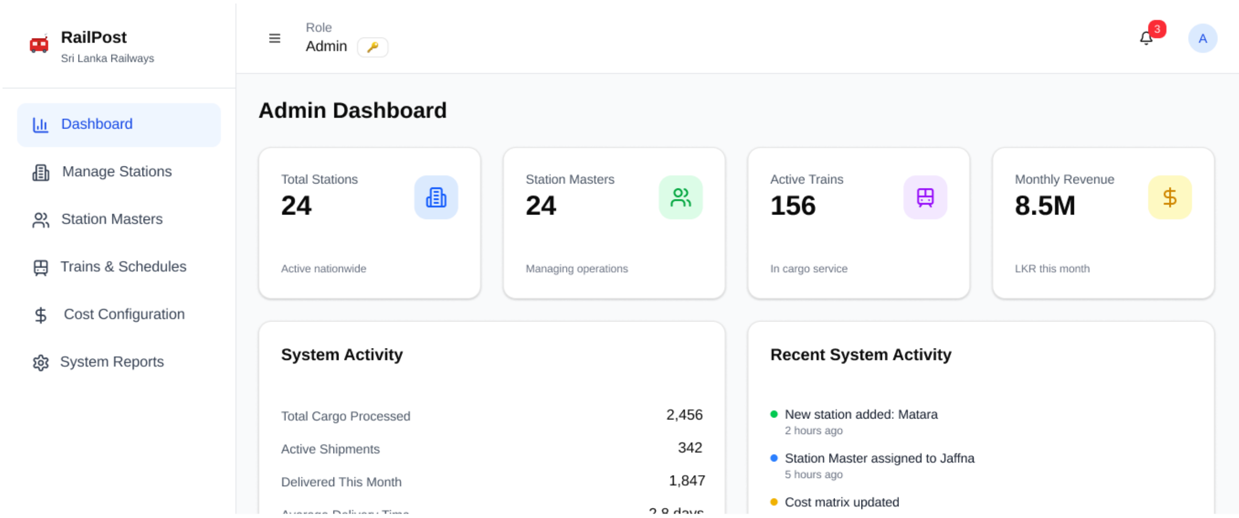


figure 04 -Sender Dashboard

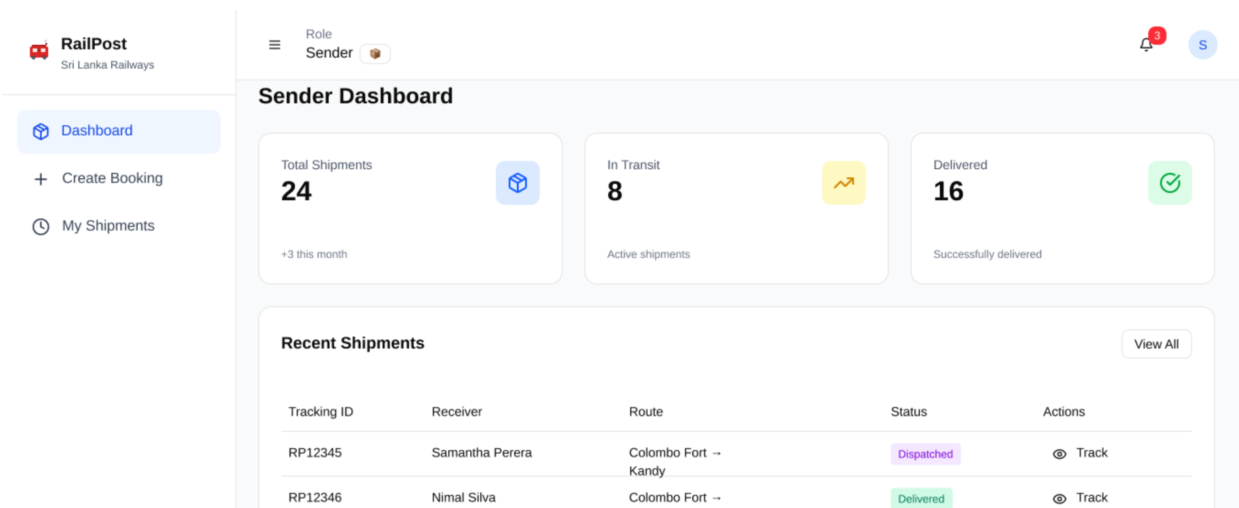


Figure 05 –Station Master Dashboard

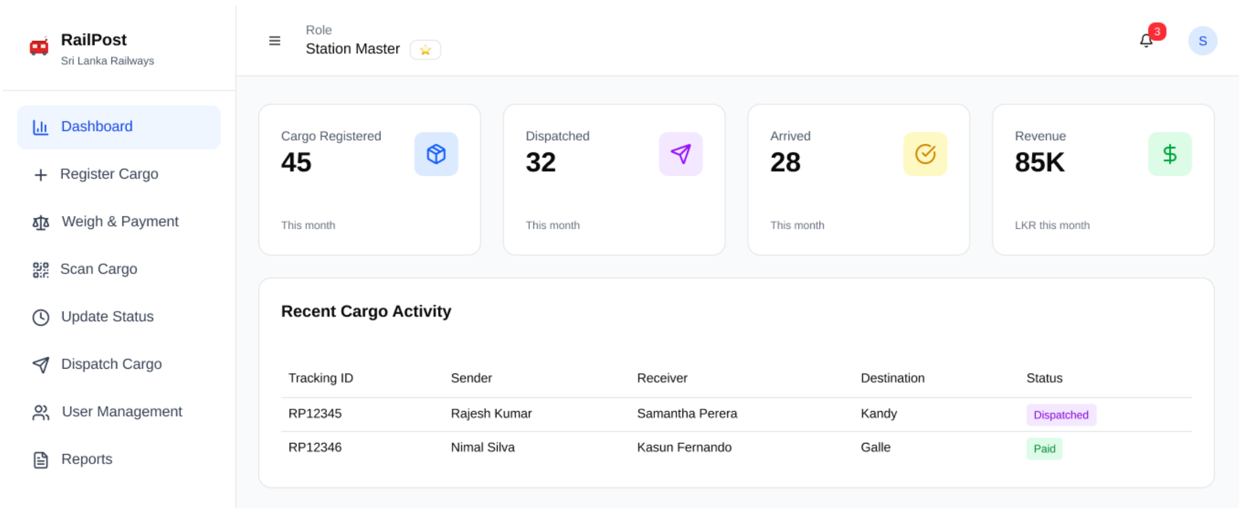


Figure 06 –Station Officer Dashboard

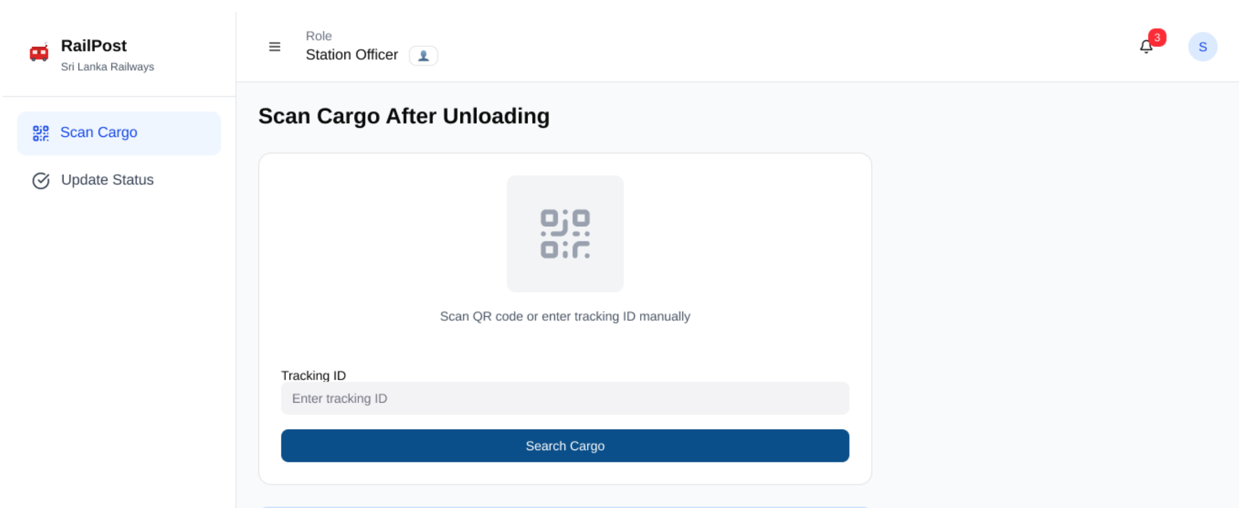


Figure 07 –Reviewer Dashboard